

Uncovering the Hidden Data Costs of Mobile YouTube Video Ads

The paper presents an empirical analysis of mobile data costs of YouTube ads. The motivation of the paper is urge video service providers to minimize data wastage linked to ads and make their services more affordable and inclusive. The ads consume a substantial amount of data, translating to direct financial cost to the user. Their findings reveal that users end up paying for 80-100% of skippable ads 31% of times in developing regions and 52.6% of times in developed regions. The key contributions of the paper include devising an experimental methodology for systematically collecting and cleaning large corpus of video ad data, conducting an in depth video buffer analysis, conducting a systematic affordability analysis to quantify the affordability of video streaming with in stream ads across all countries, and implications for different stakeholders.

The authors collect videos from the YouTube trending page, as well as the videos from the YouTube home page which are not on the trending page. Authors used two web scrappers, first to collect video urls and the second to stream the collected videos in different regions and collect the data of interest. The authors throttled the bandwidth to the average mobile bandwidth for each country in their sample. User agent string was set to a mobile device and each streamed URL was provided with the same environment and Chrome version for fair and unbiased cross country analysis. All videos were streamed at a default resolution of 360p. Data was collected by enabling stats for nerds interface.

The experimental results show a notably higher ad data proportion associated with trending videos as compared to non trending videos, which is an unexpected result given that the dataset for trending videos had fewer ads. The paper attributes it to the differences in average length of trending and non trending videos. Likewise, more ads are seen in trending videos, same is the case with developed regions. Excessive buffering of ads beyond the skip time results in buffer loss when user skips the ad. Skip loss contributes to up to 6 percent of total data consumption on the main video. Most of the skippable ads are positioned at the beginning of the video.

The paper analyzes the hidden cost of midroll ads as well. Similar to skip loss, midroll buffer loss contributes to nearly 6% of the total data consumption. Mid-roll ads in the first quarter result in a loss and subsequent re-downloading of approximately 71 seconds of the main-video content, about 58 seconds for the second quarter and 43 and 34 seconds for the remaining two quarters. With buffer loss taken into account, the ads contribute to up to 12% of the data consumed. On average, this accounts to 4.7x increase in the ad data portion. The YouTube Android app suffers from similar losses. Depending on the streaming quality, the amount of data lost due to buffer losses can be fairly significant. With all of these losses taken into account, the data plan costs can increase by up to 23.5%.

The paper proposes utilizing double ads, which may hinder the user's experience, for reducing video interruptions while maintaining the same number of ads. The authors present suggestions for streaming platforms for improving affordability and inclusiveness. The first suggestion posed in the paper is the introduction of smart media player applications that maintain multiple buffers for ads and video. Another suggestion is to prevent prefetching of skippable ad portion to avoid buffer loss.

While the work analyzes the YouTube video ads in great detail, a couple of improvement points still exist. Firstly, the experiments were conducted on Ubuntu, however, memory management differences between operating systems and browsers could have an impact on buffer loss. Repeating some of the experiments on Windows or Firefox for example, could weed out the impact of operating system or browser on the buffer management. Likewise, the paper mentions that similar results were observed on Android but does not delve into the methodology for it. Mobile devices are a popular medium of video content consumption and the impact of mobile optimizations could be further explored.

Analyzing Ad Exposure and Content in Child-Oriented Videos on YouTube

In this paper, the authors explore how advertisement exposure and content vary across child-oriented videos on YouTube, specifically in the “made for kids” labelled content. The authors show that the safety of a child’s YouTube experience is shaped significantly by their external environment. The presence of unlabelled child-oriented content with weak ad regulation indicates lax enforcement of YouTube ad and child protection policies. The paper explores 750 made for kids videos in both high policy and low policy regions. High policy countries are considered safer online environments for children with stricter child advertising policies. Developed countries tend to show more ads per video, albeit being 15% shorter in length on average.

The authors compare these 750 videos with another set of 759 unlabelled YouTube videos. The paper presents the first large scale analysis of YouTube ads on child oriented video, defines a comprehensive coding scheme to classify the age appropriateness of videos and provides policy and implementation recommendations.

In addition to using the videos from the “made for kids” section, the authors also utilize an unlabelled data set; curated using search words reflecting popular child interest. In addition, metadata, keywords and tags are analyzed for kids-directed keywords. The authors also confirm the absence of certain themes for kid oriented videos. Using Child Online Safety Index 2022, the authors analyze the policies of high ranking and low ranking countries and collect all the different types of ads using Selenium. The authors also tag each video using a codebook developed from YouTube’s stipulated policies. A video is given a primary tag (Child directed, Irrelevant, Inappropriate or Ambiguous) and some secondary tags to classify the video.

The authors’ analysis shows that ad frequency varies across regions, and the ad duration is lower in high policy regions. Analysis of YouTube ad policy compliance shows that current advertising practices deviate from YouTube Kids’ safety measures. In high policy regions, a majority of ads on YouTube kids comply with the stipulated length limits, however, the compliance is considerably lower in low policy regions. In contrast, unskippable ad limit of 15 seconds was violated across the board. Analysis of ad themes based on primary tags shows that irrelevant ads are prevalent in both regions. However, child directed ads are relatively more common in high policy regions. Inappropriate ads make up a third of the ads in low policy regions but do seldomly appear in high policy regions too. Similar results were seen for analysis based on secondary tags.

The paper also presents ad comparison across label types. The videos in the unlabelled dataset were found to thematically match the nature of videos required to be labelled as “for kids”. There is also a nearly 25% increase in the number of ads per video in the unlabelled dataset. The “made for kids” label was also found to have a limited impact on the length of unskippable ads. The label’s role in limiting ad exposure to children is currently underutilized. In both low and high policy regions, there is a marked increase in inappropriate ads when viewing child oriented content not marked as “for kids”. The authors also argue that using the search terms to casually pick a video puts the viewer at risk of exposure to unlabelled and less regulated content.

The paper raises attention towards a pressing but overlooked issue of adult and inappropriate themes in child related content on YouTube. Working with YouTube videos for the course project, I have personally experienced cases of extremely popular videos that contained “glimpses” (half a second flash for instance) of nudity or other inappropriate content. This shows that YouTube’s algorithms are either inefficient, or the platform does not monitor all videos for adhering to their own policies and guidelines. For child specific content in particular, the paper highlights the need for stricter enforcement of policies which YouTube is currently lacking. A starting step would be auto-labelling of child specific content that are not labelled by the uploader and strict scrutiny of ads displayed on the said content. If Cartoon Network can survive without showing adult themed ads on cable TV, it is safe to assume that YouTube can too.